

Separable Differential Equations

ANSWER KEY



Solving by separation of variables

- 1 Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
 $y dy = x dx$, so $\frac{y^2}{2} = \frac{x^2}{2} + C$. Using $y(0) = 3$: $C = \frac{9}{2}$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (taking the positive root since $y(0) = 3 > 0$).
- 2 Solve $\frac{dy}{dx} = 2xy$, given $y(0) = 5$.
 $\frac{1}{y} dy = 2x dx$, so $\ln|y| = x^2 + C$. Using $y(0) = 5$: $C = \ln 5$. Solution: $y = 5e^{x^2}$.

Growth, decay, and applications

- 3 A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.
 - a Solve for $P(t)$. $P(t) = 2000e^{0.05t}$.
 - b Find $P(20)$, to the nearest whole number. $P(20) = 2000e^1 \approx 5437$.
- 4 Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a 20°C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.
Solving: $T(t) = 20 + 80e^{-kt}$. Using $T(5) = 60$: $40 = 80e^{-5k}$, so $e^{-5k} = \frac{1}{2}$, giving $k = \frac{\ln 2}{5} \approx 0.1386$. So $T(t) = 20 + 80e^{-0.1386t}$.

More separable equations

- 5 Solve $\frac{dy}{dx} = \frac{y}{x}$, given $y(1) = 4$.
 $\frac{1}{y} dy = \frac{1}{x} dx$: $\ln|y| = \ln|x| + C$, so $y = Ax$ for constant A . Using $y(1) = 4$: $A = 4$: $y = 4x$.
- 6 Solve $\frac{dy}{dx} = y(1 - y)$ (logistic-type equation), given $y(0) = \frac{1}{2}$, using partial fractions $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.
 $\int \left(\frac{1}{y} + \frac{1}{1-y}\right) dy = \int dx$: $\ln|y| - \ln|1-y| = x + C$, i.e. $\ln \left| \frac{y}{1-y} \right| = x + C$. Using $y(0) = \frac{1}{2}$: $\frac{y}{1-y} = 1$ at $x = 0$, so $C = 0$: $\frac{y}{1-y} = e^x$, giving $y = \frac{e^x}{1+e^x}$.

Mixing and applied models

- 7 A tank has 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution drains at the same rate, so the volume stays constant. Let $y(t)$ be the amount of salt (kg) at time t . The differential equation is $\frac{dy}{dt} = 6 - \frac{3y}{100}$.
- a Find the equilibrium value of y (where $\frac{dy}{dt} = 0$). $6 = \frac{3y}{100}$ gives $y = 200$ kg.
- b Explain, in words, what this equilibrium represents physically. In the long run, the salt content approaches 200 kg, at which point the rate of salt entering equals the rate leaving, keeping the amount constant.

Slope fields and qualitative behavior

- 8 For the differential equation $\frac{dy}{dx} = x - y$, find the slope of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$.
At $(0, 0)$: slope $= 0 - 0 = 0$. At $(1, 0)$: slope $= 1 - 0 = 1$. At $(1, 1)$: slope $= 1 - 1 = 0$.
- 9 Verify that $y = x - 1 + 2e^{-x}$ is a solution to $\frac{dy}{dx} = x - y$, by substituting into both sides.
 $\frac{dy}{dx} = 1 - 2e^{-x}$. Right side: $x - y = x - (x - 1 + 2e^{-x}) = 1 - 2e^{-x}$. Both sides equal $1 - 2e^{-x}$, confirming the solution.

Visualizing a logistic solution curve

- 10 The graph shows the logistic solution $y = \frac{e^x}{1 + e^x}$ found earlier for $\frac{dy}{dx} = y(1 - y)$, $y(0) = \frac{1}{2}$. Use the graph to describe the long-run behavior of y as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.
As $x \rightarrow \infty$, $y \rightarrow 1$ (the curve levels off approaching the horizontal asymptote $y = 1$). As $x \rightarrow -\infty$, $y \rightarrow 0$.
This matches the equilibrium solutions of $y(1 - y) = 0$ at $y = 0$ and $y = 1$.

Another applied separable equation

- 11 Radioactive decay follows $\frac{dA}{dt} = -kA$. If a sample decays from 80 g to 20 g in 30 days, find k and the half-life of the substance.
 $A(t) = 80e^{-kt}$. Using $A(30) = 20$: $\frac{20}{80} = \frac{1}{4} = e^{-30k}$, so $k = \frac{\ln 4}{30} \approx 0.0462/\text{day}$. Half-life: $e^{-kt_{1/2}} = \frac{1}{2}$ gives $t_{1/2} = \frac{\ln 2}{k} = \frac{30 \ln 2}{\ln 4} = 15$ days.

Separable equations with algebraic solutions

- 12 Solve $\frac{dy}{dx} = \frac{x^2}{y^2}$, given $y(0) = 2$.
 $y^2 dy = x^2 dx$: $\frac{y^3}{3} = \frac{x^3}{3} + C$. Using $y(0) = 2$: $\frac{8}{3} = C$. Solution: $y^3 = x^3 + 8$, i.e. $y = (x^3 + 8)^{1/3}$.
- 13 Solve $\frac{dy}{dx} = \frac{1}{2y}$, given $y(0) = 3$.
 $2y dy = dx$: $y^2 = x + C$. Using $y(0) = 3$: $C = 9$. Solution: $y = \sqrt{x + 9}$ (positive root, since $y(0) = 3 > 0$).

Verifying a solution to an applied model

- 14** For the mixing problem $\frac{dy}{dt} = 6 - \frac{3y}{100}$ with $y(0) = 0$, verify that $y(t) = 200(1 - e^{-0.03t})$ is a solution, and confirm it approaches the equilibrium 200 found earlier as $t \rightarrow \infty$.
 $\frac{dy}{dt} = 200(0.03)e^{-0.03t} = 6e^{-0.03t}$. Right side:
 $6 - \frac{3}{100} \cdot 200(1 - e^{-0.03t}) = 6 - 6(1 - e^{-0.03t}) = 6e^{-0.03t}$. Both sides match. As $t \rightarrow \infty$, $e^{-0.03t} \rightarrow 0$, so $y(t) \rightarrow 200$, confirming the equilibrium.
- 15** Solve $\frac{dy}{dx} = e^{x-y}$, given $y(0) = 0$, using the fact that $e^{x-y} = e^x \cdot e^{-y}$.
 $e^y dy = e^x dx$: $e^y = e^x + C$. Using $y(0) = 0$: $1 = 1 + C$, so $C = 0$: $e^y = e^x$, giving $y = x$.
- 16** A pond's fish population P grows according to $\frac{dP}{dt} = 0.08P(1 - \frac{P}{500})$, a logistic model with carrying capacity 500. Without solving the equation explicitly, determine the equilibrium population levels (where $\frac{dP}{dt} = 0$), and state which one is realistic for a growing population starting from $P(0) = 50$.
Setting $\frac{dP}{dt} = 0$: $P = 0$ or $P = 500$. Starting from $P(0) = 50 > 0$, the population will grow toward the stable equilibrium $P = 500$ (the carrying capacity), since $P = 0$ is an unstable equilibrium for any positive starting population.
- 17** Solve $\frac{dy}{dx} = \frac{\cos x}{y}$, given $y(0) = -2$, and state why the negative root must be chosen.
 $y dy = \cos x dx$: $\frac{y^2}{2} = \sin x + C$. Using $y(0) = -2$: $2 = 0 + C$, so $C = 2$: $y^2 = 2\sin x + 4$, giving $y = -\sqrt{2\sin x + 4}$. The negative root is required because $y(0) = -2 < 0$, and y must vary continuously from that starting sign.

A final applied separable model

- 18** A drug is eliminated from the bloodstream according to $\frac{dC}{dt} = -0.2C$, where C is concentration (mg/L) and t is hours. If the initial dose gives $C(0) = 50$ mg/L, find $C(t)$, then find how many hours it takes for the concentration to drop below 5 mg/L, to one decimal place.
 $C(t) = 50e^{-0.2t}$. Setting $50e^{-0.2t} = 5$: $e^{-0.2t} = 0.1$, so $t = \frac{\ln 0.1}{-0.2} = \frac{-2.3026}{-0.2} \approx 11.5$ hours.